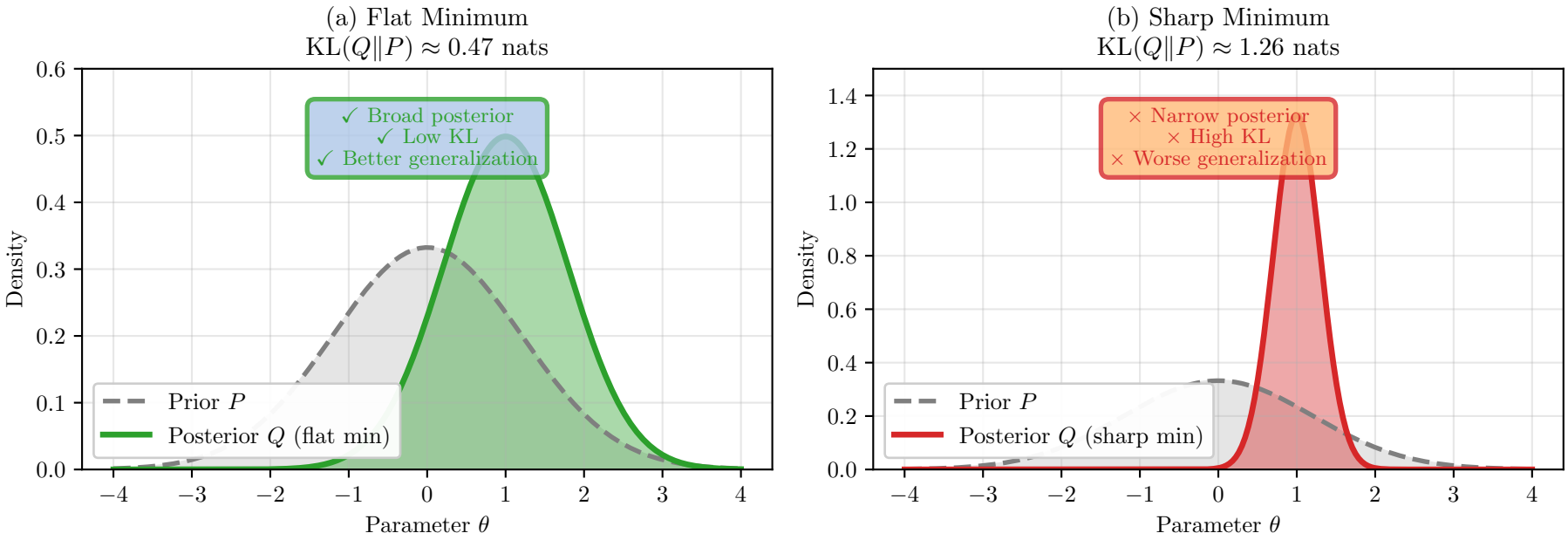
Flatness ↔ Low KL ↔ Good Generalization
(a) Flat Minimum
KL(Q‖P) ≈ 0.47 nats
✓ Broad posterior
✓ Low KL
✓ Better generalization
Prior P
Posterior Q (flat min)
Density
Parameter θ
(b) Sharp Minimum
KL(Q‖P) ≈ 1.26 nats
× Narrow posterior
× High KL
× Worse generalization
Prior P
Posterior Q (sharp min)
Density
Parameter θ